Internship Report

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Title: High Efficiency Irrigation Systems

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Ali Akbar Group

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1 Introduction

Ali Akbar Group is a Pakistan based agriculture company marketing and manufacturing seeds, pesticides, fertilizers, high efficiency irrigation solutions. Target Zari Markaz and DJC shops are premier Franchise and dealership networks available all across Pakistan. Huge field advisory service team, state of the art laboratory, finest manufacturing and formulations facilities provide this unique relation of trust of millions of farmers in Pakistan. The foundation of the group is laid on providing excellence towards transformation in agricultural systems; in particular enhancing farmers' yields and sustainable economies, development with an objective to enhance the quality of human life. The report is based on study of **High Efficiency Irrigation Systems**

2 HEIS: High Efficiency Irrigation Systems

A high efficiency irrigation system is a modern and advanced method of watering crops and plants that maximizes water usage while minimizing waste. This innovative system utilizes cutting-edge technology and smart design to deliver water precisely to the roots of plants, reducing evaporation and runoff. The key components of this system often include drip irrigation, or precision sprinklers that release water slowly and directly to the plant's base. Additionally, some systems are equipped with weather-based controllers, soil moisture sensors, and remote monitoring capabilities to ensure that water is applied only when and where it is needed most. By optimizing water distribution and minimizing water loss, high efficiency irrigation systems not only conserve water resources but also promote healthier plants and increased crop yields, making them a sustainable and environmentally responsible choice for agriculture and landscaping practices. Drip and Sprinkler systems will be discussed in detail for this report.

2.1 Types Of Systems

High efficiency irrigation systems comprise of the following:

2.1.1 Drip Irrigation

Drip irrigation is a highly efficient and precise method of watering plants and crops, designed to deliver water directly to the roots in a slow, steady, and controlled manner. This irrigation system operates by using a network of tubes, pipes, and emitters to distribute water directly at or near the plant's base, minimizing water wastage through evaporation or runoff. The

emitters, commonly in the form of drip emitters or micro-sprinklers, release water drop by drop, ensuring that plants receive a consistent and uniform supply of moisture. By optimizing water distribution and minimizing water loss, high efficiency irrigation systems not only conserve water resources but also promote healthier plants and increased crop yields, making them a sustainable and environmentally responsible choice for agriculture and landscaping practices.



Figure 1: Drip line with laterals

2.1.2 Sprinkler System

A sprinkler system is a popular and efficient method of irrigating lawns, gardens, and agricultural fields. It consists of a network of pipes, valves, and sprinkler heads that distribute water over a designated area in a controlled and systematic manner. Sprinkler systems come in various types, including oscillating, impact, rotor, and pop-up sprinklers, each with its unique pattern and range of coverage. When the system is activated, water is released from the sprinkler heads in a spray or mist, effectively simulating natural rainfall. The versatility of sprinkler systems allows for customization to suit specific watering needs, such as adjusting the spray distance and pattern to accommodate various plant types and soil conditions. Furthermore, modern sprinkler systems can be integrated with weather-based controllers and moisture sensors, enabling automatic adjustments based on real-time weather data and soil moisture levels. While sprinkler systems can be effective in maintaining lush landscapes and agricultural productivity, proper design, maintenance, and scheduling are essential to ensure water efficiency and prevent wasteful practices.



Figure 2: Impact Sprinkler

RainGun

A rain gun, also known as a water cannon or water gun, is a specialized irrigation tool used for large-scale agricultural and landscape irrigation. It is a type of impact sprinkler that is designed to cover vast areas with water in a circular or sector pattern. The rain gun operates by propelling a high-pressure stream of water through a nozzle, which breaks the water into droplets that disperse over the intended area. Due to their high water output, rain guns can quickly cover substantial distances, reducing the overall irrigation time and labor required. However, it is crucial to use rain guns judiciously and efficiently to prevent water wastage and ensure that water reaches the target area without excessive evaporation or runoff.



Figure 3: Rain Gun

Central Pivot System

A central pivot system, also known as a center pivot irrigation system, is a highly efficient and widely used method of irrigating large agricultural fields. This innovative system consists of a series of towers that support a long, rotating pipeline, with water being pumped from a central source. As the pipeline pivots around the central point, water is evenly distributed through sprinklers or drip lines, covering a circular area of crops. The central pivot system

is beneficial for farmers as it minimizes water wastage, reduces labor requirements, and maximizes crop yield by providing consistent and precise irrigation.



Figure 4: Central Pivot system

2.2 Design Considerations

Design of an irrigation system involves following steps.

1. Area to Be irrigated
2. Type of Crop
3. Crop Water Requirement
4. Estimate of Materials
5. Water Source Selection
6. Pump Selection
7. Cost of operation

2.3 Components and fittings

2.3.1 PVC Pipe

PVC pipes play a vital role in modern irrigation systems. These durable and cost-effective pipes are widely used for transporting water to various parts of the farm or garden.



Figure 5: PVC pipes

2.3.2 LDPE

LDP (Low-Density Polyethylene) pipe is a type of flexible plastic pipe commonly used in irrigation and water distribution systems. LDP pipe is particularly suitable for low-pressure applications and is commonly used in agriculture for drip irrigation and sprinkler systems. Its affordability and adaptability have made it a popular choice for small-scale irrigation projects and landscaping applications.



Figure 6: LDPE pipe

2.3.3 HDPE

HDPE pipes are often used as the mainlines in sprinkler irrigation systems. The flexibility of HDPE pipes allows for easy maneuvering around obstacles, ensuring proper water distribution to the crops.

2.3.4 Elbows

Elbows are most commonly available in 90-degree angles and 45-degree angles, so your pipeline can be directed however you want it. Side elbows are a slightly different piece that are commonly used as corners in projects such as canopies, tents, displays, and boxes. They have three ends instead of two. Most elbows have slip socket ends, but threaded versions are available.



Figure 7: 4” Elbow

2.3.5 Tees

Tees allow a line to be split into two separate lines with a 90-degree connection. Also, tees can connect two lines into one main line. They are also often used for PVC structures. Tees are an extremely versatile fitting that are some of the most widely used parts in plumbing. Most tees have slip socket ends, but threaded versions are available.



Figure 8: 4" Tee Fitting

2.3.6 Crosses

type of fitting that join four pipe sections. They have four slip connections that meet at 90-degree angles, forming a "plus" shape. PVC cross fittings are usually used when building framework out of PVC pipe. Crosses can add a great deal of structural integrity to a framework. They can also be used to divide fluid flow in different directions.



Figure 9: Cross Tee

2.3.7 Couplings and Unions

Couplings are one of the most simple (and inexpensive) types of PVC fittings. They are a small part that connects or "couples" one part to another, usually permanently. They can connect pipe to pipe and pipe to fitting. Some of them even reduce so you can connect a small pipe to a large pipe or vice versa. Couplings are available with slip or female-threaded ends.

Unions, like couplings, connect things, but they are a not-as-permanent solution that can be easily taken apart. PVC unions are often used in building temporary structures, such

as tent supports, and can be taken apart when the structure is no longer needed. They feature a ring in the center that separates the two ends from each other, allowing for easy deconstruction and maintenance.



Figure 10: Coupling and Union

2.3.8 Caps and plugs

Caps have a very simple job: stop the flow. They are put on the end of a pipeline that does not need to be connected to another pipe. Plugs are like caps, but instead of stopping the flow in a pipe, they stop the flow in a fitting.

2.3.9 Flanges

Flanges are a disc-like fitting that create a tight seal by pressing two surfaces together with bolts, clamps, edges, or a compressive force. Most flanges, however, use bolts to join the two surfaces together. PVC flanges are available with slip or threaded ends.

2.3.10 Adapters

Adapters, sometimes called reducing couplings, are a versatile fitting. These fittings are designed to change the end type of a pipe, allowing it to connect to fittings and pipes of many sizes. They can have threaded or slip socket ends to connect to an endless variety of pipes and fittings.

2.4 Design of system

Designing of Drip system can be understood by the following examples:

> **Example 1**: A drip system is to be installed on a farm site of 1320×985 ft, plant and row spacing are both 20ft and the no.of drippers for each plant is 6. Find the Area in acres,No.of plants,Total drippers and Length of dripline.

Solution:

$$\underline{\text{Area}} = 1300200 \text{ Ft}^{\,2}$$

OR

29.84 Acre

$$\text{No.of Plants} = \frac{Area}{(Row--Row)(Plant--Plant)}$$

$$\underline{\textbf{No.of Plants}} = \frac{130200Ft^2}{(20Ft)(20Ft)} = 3251$$

There are 6 drippers for each plant:
Thus, the total **Number of Drippers** are $3251 \times 6 = 19506$
Length of Drip-Line required:

$$\frac{Area}{(Row--Row)}$$

Putting the values,

$$\frac{130200Ft^2}{(20Ft)} = 65010Ft$$

OR

$$19816m$$

For a double dripline: **Length required** $= 130,020Ft$
OR

$$39632m$$

Example 2: For the example given above find the number of rolls of dripline,Total flow of the entire system, Required H.P for the pump and power requirement for the motor. How can pump selection be made?

Solution:
Assuming one roll= 400m
Then,

$$\textbf{No.of rolls required} = \frac{39632m}{400m} = 100$$

$$\text{Total Flow} = 16 \times 19506 = 312096\text{LPH OR } \textbf{86 LPS}$$

$$\text{Pump H.P} = \frac{Q \times H}{75 \times \eta_p \times \eta_w}$$

where, Q= Flow rate in LPS
H= Head (assumed 40m)
η_p= pump efficiency (85%)
η_w= Water efficiency (60%)

Putting Corresponding values: Pump H.P= 90hp

As a thumb rule the required power for solar system is the same as for the pump so a solar system of 90 kw output or above is required

Motor Power required:

$$1\text{H.P} = 746W$$

$$90\text{H.P} = 746 \times 90$$

$$90\text{H.P} = 62,140W$$

OR

$$\text{Motor-Power} = \frac{67140}{1000} = 67.14KW$$

The pump selection can be made based on H-Q pump curves.

The size of pipe required for this flow can be roughly estimated by Flow = (Size $\times$ 2) + 1 which yields about 43" pipe.

A pipe of diameter this size is impractical for irrigation purposes thus, the area should be bifurcated since it is not necessary to irrigate entire area at once.

The area can be divided into 10 zones i.e $\frac{29.84}{10} = 2.9acre$. In this case;

$$\text{No. of Plants} = 325$$

$$\text{No.of drippers} = 1951$$

$$\text{Length of drpline} = 3963.2$$

$$\text{Flow} = 8.6\text{LPS}$$

Similarly, H.P required will be 9 units and solar power required will be 9 Kw. Now, a 4" line may be used to irrigate a zone.

Example 3: The irrigation system installed has an efficiency of 90%. The reference evapo-transpiration has a value of 7.2. Taking crop coefficient as 1.1 and canopy factor of 58%, Find peak daily consumption,Application Rate and operational time for the system. Take other values from examples above.

Solution:

$$\text{Peak daily consumption} = \frac{ET^{\circ} \times K_c \times C.F}{\eta}$$

$$\text{Peak daily consumption} = \frac{7.2 \times 1.10 \times 0.58}{0.9} = 5.104\frac{mm}{day}$$

$$\text{Application Rate(A.R)} = \frac{\text{Flow of zone}}{\text{Area of Zone}} = \frac{31320\text{LPH}}{11736m^2} = 2.668\frac{mm}{hr}$$

$$\text{Operation Time} = \frac{PeakDaily}{A.R} = \frac{5.104}{2.668} = 1.9\frac{hr}{day}$$

The calculation is based on one zone of 2.9 acre, thus the operational time for 10 zones is 19 hours which is impractical. So application rate of the dripper will be increased such that a 24 LPH dripper may be used instead of 16LPH.

3 Site Visiting And activities

3.1 Javed Organic Farm

Located on Sargodha-Mandibahaudin road. It is an area of 10 acres for citrus growth. 3 Acre of the land is being used for lemon while 7 acres for orange.
Plant to plant distance is kept at 20ft and bifurcated into 5 zones of 2 acres each.

A Drip System is being used to irrigate the citrus orchard. A head unit is installed whose motor is powered by solar system. The water is sourced from a pond.



Figure 11: Drip and emitter



Figure 12: Head Unit

3.2 New Metro City-Mandi Bahauddin

The key objectives of the activity were as follows:

1. Participate in the installation of popup spray heads in the centrimedian of New Metro City housing society road.
2. Observe the planning and execution process for an efficient irrigation system.
3. Understand the significance of water conservation in urban landscape management.

3.2.1 Methodology

The installation of popup spray heads involved the following steps:

1. Site Assessment: A thorough assessment of the centrimedium was conducted to determine suitable positions for the popup spray heads.
2. Excavation: Specialized tools were used to excavate trenches for laying the irrigation pipelines. .
3. Pipeline Installation: PVC pipes were joined such that the diameter of the pipe was reduced as it moved away from the pump (by means of a reducer) in order to compensate for pressure head loss so flow can be balanced.
4. Holes were drilled onto the main line; rubber was adjusted in the hole and GTO(Gromate take-off) was inserted into it.
5. LDP(Low Density poly-ethylene) was adjusted onto GTO which is used to connect popup inlet to the water line.
6. Testing: Once the installation was complete, the irrigation system was thoroughly tested to ensure that the popup spray heads functioned correctly and distributed water evenly. Moreover, it was made sure that no dry patches are left.



Figure 13: Drilling on Main line



Figure 14: Joining Pipe



Figure 15: LDP connection to Main Line



Figure 16: Intalled Spray-head

3.2.2 Results

1. Water Conservation: The use of advanced popup spray heads ensured optimal water usage, promoting water conservation within the housing society.

2. Operation: The irrigation system was fully functional as water was distributed evenly and no dry patches were observed. All the popup heads were operating under the pressure delivered. Flushing of the system was done by means of a flush valve prior to starting of system.

3.2.3 Specifications

Table 1: Specifications of materials

Item	Quantity	Specs
Popup Spray head(SS-530)	50	Flow: 1.3 GPM, radius: 9'-19', Min.pressure: 3bar
PVC pipe		4",2" and 1.5"
LDPE	50 short piece	16mm
GTO,Rubber	50	16mm
Rathcet clamp,L.adaptor	50	16mm

3.2.4 Notes

1. The Underground pipes were crossed using sleeves.
2. The rise of the popup spray head at operational pressure is 4".
3. Spacing of 15 ft was used for Popup spray heads.
4. Thus, a hole at every 15ft of the centrimedian
5. Spray heads were installed diagonally because the throw from the head to the front is lesser as compared to sides.
6. 1 foot deep trench was dug for main-line.

3.3 Raja Atah Ullah Farm

It is a Impact sprinkler site located near Dharyala Jhalip in district Jehlum, Tehsil P.D Khan.

The installation area spans 12.1 acres along the length of 1300 ft of 4" main supply line. A total of 40 Impact sprinklers having coverage of 50ft or 15m are installed at the site. The water is sourced from a pre-filled pond which in turn is replenished by ground water. The water is taken up by pump powered by a diesel engine of 16hp.

3.3.1 Methodology

The methodology for installation is the same as above however a single size of pipeline is used throughout the area since, the all sprinklers will not operate at same time eliminating a significant pressure loss. HDP (High density poly-ethylene) pipes are run perpendicular to the main line.



Figure 17: Head Unit-Power source



Figure 18: Main Line (1300ft)

The trench dug for the main line is 2.5ft deep and valves are present at every 50m along its length. these valves are installed by cutting the pipe and joining it by the means of a Tee (PVC fitting) which reduces the 4" line to a 2" pipe using a pre-built reducer so a connection from 2.5ft can be made to the ground level. Ball valves are placed on 2" pipe at the ground level through which HDP pipes are run perpendicular.
The water runs through the HDP pipe to a 1" riser pipe towards the sprinkler.



Figure 19: Cutting Pipe



Figure 20: Installed Tee



Figure 21: Cut length measurement

3.3.2 Specifications

The specifications of the HEIS can be seen in the Figures below:

Item Description	Imported - ISO Certified/ Local	Make/ Model	Standard & Specification	Unit	Quantity	Unit Rate (Rs.)	Cost (Rs.)	GST/ PST Rate %	GST/ PST Unit Rate (Rs.)	Total GST/ PST (Rs.)	Cost incl. GST/ PST	Total Cost (Rs.)	Remarks
	as per already approved inventory list of HEIS equipment			a	b	c (not higher than the already approved rates of HEIS equipment)	d = b*c	e	f	g	h=d+g		
Pump Set & Prime Mover													
Surface pump KSB-40-200	local		Q =8.67 lps, H= 40 m	No			-	17	-	-	-	-	
Diesel Engine (Chief)	Local		16 hp	No			-	17	-	-	-		
BP & Coupling for motor/ engine & pump	Local		Local	No			-	17	-	-	-		
uPVC/HDPE Pipes													
uPVC Pipe	Any ISO (BS-		4"/ 110 mm (B-Class)	m			-	17	-	-	-		
uPVC Pipe	Any ISO (BS-		2"/ 63 mm (C-Class)	m			-	17	-	-	-	-	
HDPE Pipe with Latch/ Both side quick coupling 63mm	Imported		63 mm (PN-6)	m			-	17	-	-	-		
uPVC/HDPE Pipes													
uPVC Pipe	Any ISO (BS-		4"/ 110 mm (B-Class)	m			-	17	-	-	-		
uPVC Pipe	Any ISO (BS-		2"/ 63 mm (C-Class)	m			-	17	-	-	-	-	
HDPE Pipe with Latch/ Both side quick coupling 63mm	Imported		63 mm (PN-6)	m			-	17	-	-	-		
Emitting System/Laterals													
Impact Sprinklers	Imported - ISO Certified/ local		R-15m, 1.66 m^3/hr	No.			-	17	-	-	-	-	
Riser Pipe	Local		1" (3ft long)	No.			-	17	-	-	-		
Valve System and Monitoring Equipment													
PVC Ball Valve	Imported (SCH-80)		2" / 63 mm	No.			-	17	-	-	-		
PE End Stopper with Latch	Imported		2" / 63 mm	No.			-	17	-	-	-		
PE ABOT/ Sprinkler Button with Latch	Imported		2" / 63 mm	No.			-	17	-	-	-	-	
Elbow/Bend with Latch/quick Coupler	Imported		50mm	No.			-	17	-	-	-		
Flow Meter	Imported		3"	No.			-	17	-	-	-		
Pressure guages	Imported - ISO Certified		2.5" , 1-10 bar	No.			-	17	-	-	-		
Fittings and Accessories													
Fittings and Accessories													
Head unit Fittings including Suc pipe and foot valve	Lump Sum		3"	No.			-	17	-	-	-		
PVC Fittings	Imported		Sch-80	L/S		7% of PVC Pipe	-	17	-	-	-		
Handle Valve	Local		2"	No.			-	17	-	-	-	-	
Air Release Valve	local		1"	No.			-	17	-	-	-		
Non-Return Valve	local		3"	No.			-	17	-	-	-		
Others													
Signboard	As per Dept Approved Specs			No.			-	17	-	-	-		
Excavation	As per Dept Approved Specs		0.92m (3') x 0.46m (1.5') x 0.3m (1')	m			-	16	-	-	-	-	
Head Unit Foundation	As per Dept Approved Specs		2.5m x 1.5m x 0.5 m	No.			-	17	-	-	-		
Total Material Cost				Rs.			-			-	-	-	

3.3.3 layout/visual

For any installation site a layout design is provided which helps the technician in the installation process and gives an exact overview of the plan.
The design layout for this this farm can be seen in the figure below.

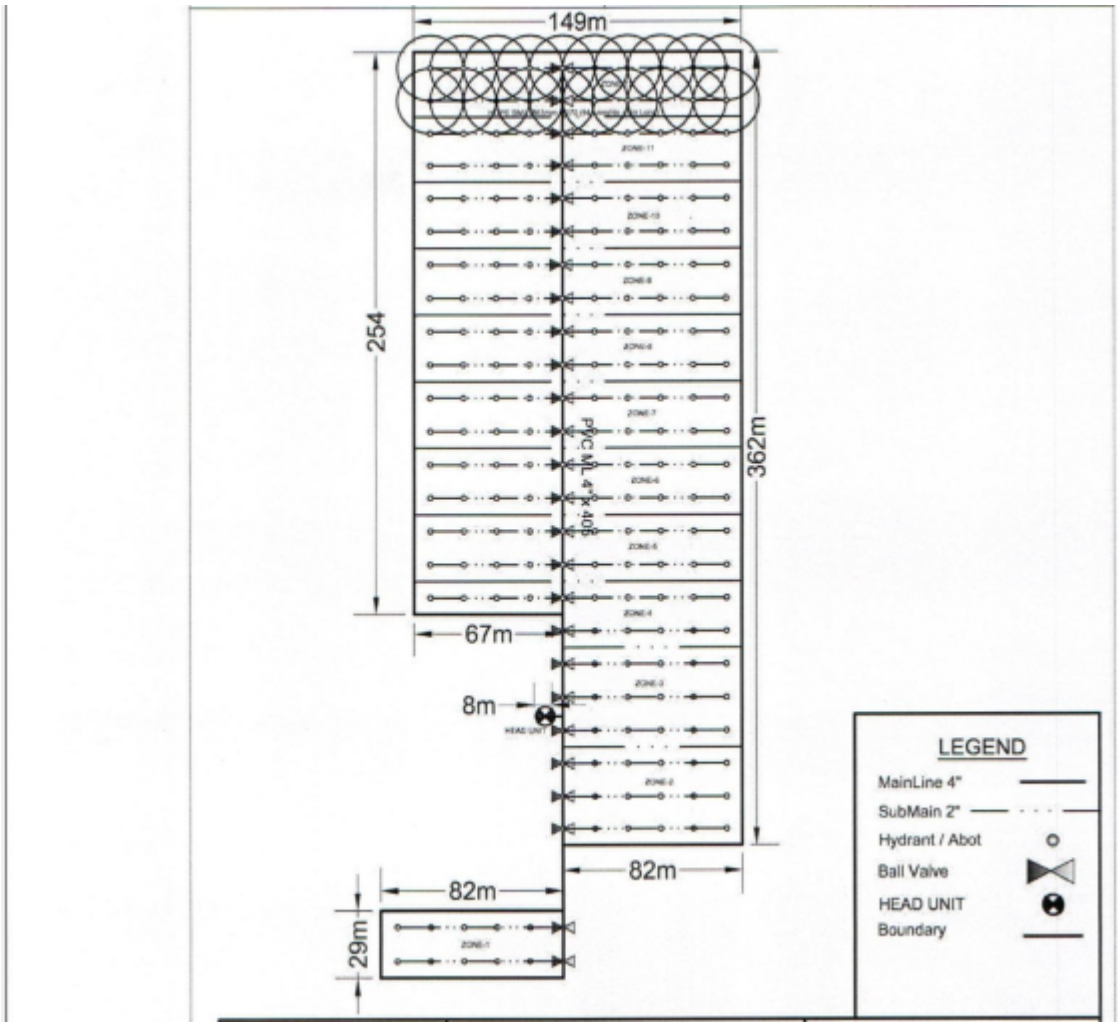


Figure 22: Design Layout of Atah Ullah's Farm

The circles in the figure represent the coverage radius for each sprinkler. It should be noted that the circles overlap which suggests the designed spacing is good.
If the spacing of sprinkler is not proper overlap my not occur which will leave dry patches in between.

4 Layout and Designing at Headquarters

The layout preparation was done on Autocad. The process involved are listed below.

- Getting coordinates of the boundary.
- Determining the location of Head unit.
- Determining Area of installation.
- Determining power requirements.
- Importing path/Boundary in Autocad.
- Finding pipe length (Main and Submain)
- Determining Head loss through Excel sheet
- Adjusting power requirement in spreadsheet.

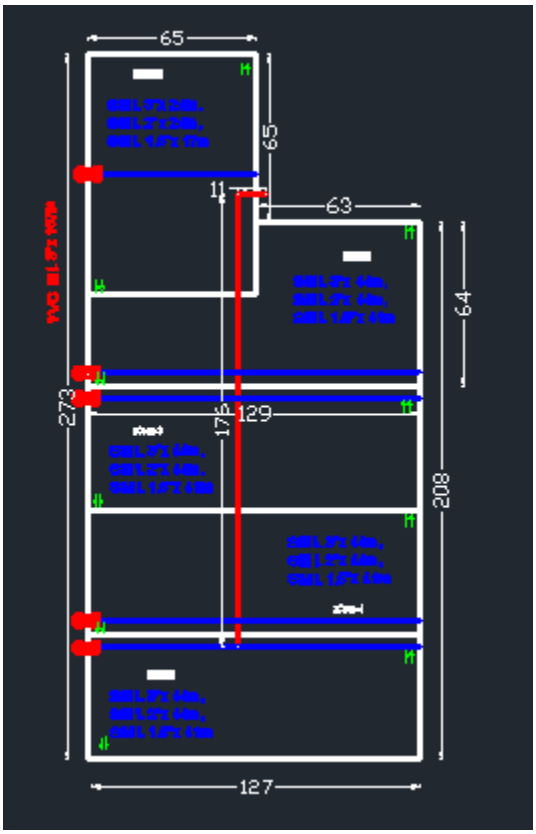


Figure 23: Drip layout in autocad

Name of SSC:Ali Akbar Enterprises pvt. Ltd.			
TYPE OF SCHEME:	**Drip Orchard**		
I) GENERAL INFORMATION			
Name of Beneficiar	**ijaz Hayyat**	Town/ Village/ mouza:	**Jhavrian**
Tehsil:	**Shakpur**	District:	**Sargodha**
Province:	**Punjab**	Phone. No:	**0333-4949493**
Land Holding (Acr		Scheme Area (acres)	**7.6**
Topography (Flat/undulate	**Flat**	N31 50.664 E72 42.323	
CNIC			**(Attach topographic map)**

Figure 24: General Information about Land

PUNJAB RESILIENT AND INCLUSIVE AGRICULTURE TRANSFORMATION (PRIAT)

Particular	Unit	Irrigation Blocks						
		Block-I	Block-II				section -II	Maximum
Lateral Length	m	66						
Lateral (inside diameter)	mm	13.6						
No. of Emitter	Nos.	36						
emitor flow rate	LPH	12.0						
Lateral flow rate	LPH	434						
Head loss in lateral	m/100m	2.35						
	m	**1.55**						**1.55**
Sub-mainline (inside diameter)	mm			82.3	54.5	42.5		
Sub-Mainline: length	m			44	44	41		
Sub-Mainline: flow rate	LPH			18436	12500	7500		
Sub-Mainline: Head Loss	m/100m			0.39	1.43	1.86		
	m			**0.17**	**0.63**	**0.76**		**1.56**
Sub-Mainline: Velocity	m/s			0.96	1.49	1.47		
Mainline: Inside diameter	mm	82.3						
Mainline: length	m	187						
Mainline: flow rate	LPH	18436						

Figure 25: Headloss Information

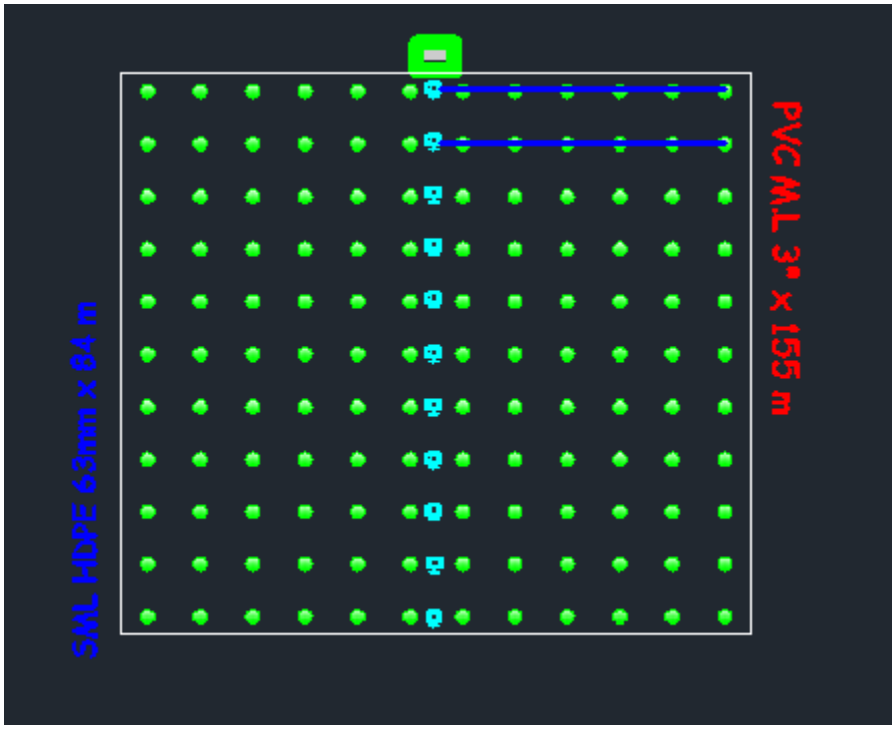


Figure 26: Sprinkler layout

	Beneficiary	Gen Yousaf							
	Area (Acres)	7.13							
	District	Sargodha							
	BILL OF QUANTITIES (SPRINKLER IRRIGATION SYSTEM)								
	Description	Make & Model&Standard	Unit	Quantity	Unit Price (Rs.)	Total Price (Rs.)	GST/PST Rate (%)	GST/PST Amount (Rs.)	Total Amount (Rs.)
A- Pumping Unit									
1	Centrifugal pumpset H = 51m, Q = 5.2 lps	Hydro 2x1.5"	Nos.	1	88,966	**88,966**	18%	16,014	**104,980**
2	Electric Motor (2900rpm)	Siemens 7.5 Hp	Nos.	1	169,586	**169,586**	18%	30,525	**200,111**
3	BP & Coupling for motor & pump	Local	Nos.	1	27,509	**27,509**	18%	4,952	**32,461**
	Sub Total: (Pumping Unit)					**286,061**		**51,491**	**337,552**
Pipes, Fittings & Trenching									
1	PVC Pipe 3 inch B-Class	BS-3505 or Equiv.	M	304	773	**234,992**	18%	42,299	**277,291**
2	PVC Pipe 2 inch C-Class	BS-3505 or Equiv.	M	24	433	**10,392**	18%	1,871	**12,263**
3	HDPE Pipe with Latch/Both side quick coupling 63mm	Imported	M	400	1,119.3	**447,720**	18%	80,590	**528,310**
4	PVC Fitting (7% of total pipe cost)	Imported	Nos.	1	17,177	**17,176.88**	18%	3,092	**20,269**

Figure 27: Bill of Quantities

Flow meter losses		m	**2**
Miscellaneous head loss		m	**3**
Pumping lift (Suction)		m	**6**
Pumping lift (Delivery)		m	
Total Dynamic Head:		m	**47**
Q LPS	**6.0**	**Head m**	**47**
Combine η %	**51**	**HP**	**7.4**
Motor η 85% & pump η 60%			**Say 7.5 Hp**

Figure 28: adjustment of hp

THANK YOU!